Introduction- Cathode Ray Experiment

The CRT experiment was used to discover the electron. In the experiment, a tube with one end containing the Cathode (-) and the other, the Anode (+). These two electrodes were connected to a high voltage power source. A vacuum pump was used to reduce the pressure considerably.

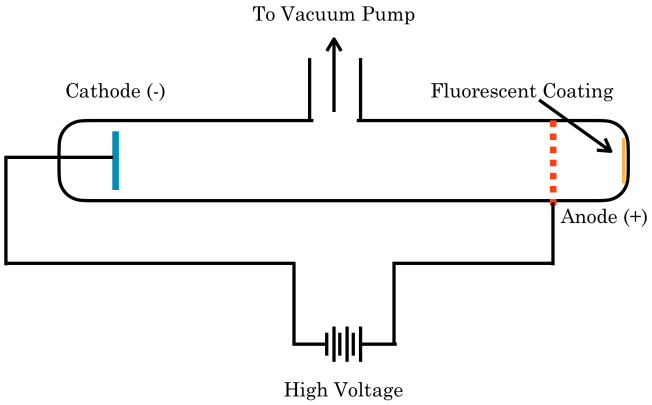


Figure 1: Cathode Ray Discharge Tube

The particles moving from the Cathode to Anode were called Cathode Rays. These were fast moving electrons. Particles moving from Anode to Cathode were called Anode Rays. These are Protons. These rays cause fluorescent material to glow. Charge to mass ratio (e/m) is a universal constant for cathode rays:

$$e/m = 1.76 \times 10^{11}\,\mathrm{C\,kg^{-1}}$$

R.A. Millikan's Oil Drop Experiment

In his experiment to calculate charge on electron, Millikan had a chamber as shown below. The Oil Droplets were sprinkled and the positively charged metal plate had a hole just large enough for a single droplet to pass through. The X-Ray source is highly ionizing and removes electrons from molecules floating in the air which go and stick to the oil droplet, giving it a negative charge. This caused it to be repelled by the negatively charged plate at the bottom.

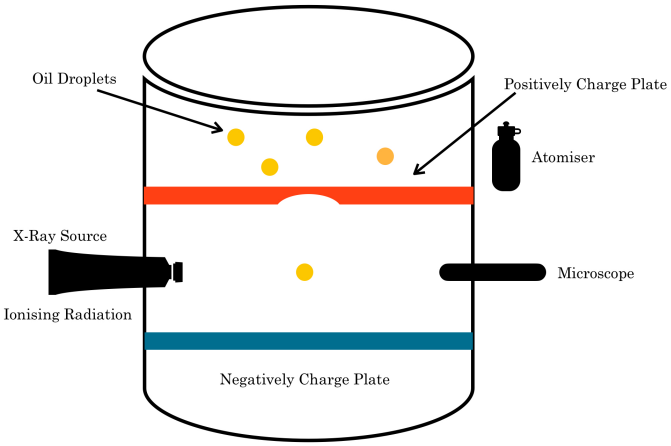


Figure 2: Oil Drop Experiment

By increasing the Electric Field of the negatively charged plate, he was able to get the oil droplet to

stay stationary in the air, using which the charge on the electron came out to be:

$$e = 1.6 \times 10^{19}\,\mathrm{C}$$

Discovery of Neutron- James Chadwick

Chadwick bombarded a thin sheet of Beryllium with α-particles and neutral particles having mass slightly greater than that of a proton were emitted.

$$^{9}_{4}\mathrm{Be} + {}^{4}_{2}\mathrm{He} \rightarrow {}^{12}_{6}\mathrm{C} + {}^{1}_{0}\mathrm{n}$$

1 Rutherford's Experiment

1.1 Gold Foil Experiment

Rutherford placed a lead box through which α-particles moved towards a gold foil roughly 1000 atoms thick. He expected that there would be deviation $< 1^{o}$. But, most particles passes straight through the gold foil, some were deflected by small angles while very few particles rebounded. (Deviation = 180°)

1.2 Nuclear Model of Atom

Rutherford proposed that most of the atom was composed of empty space, because most α-particles passed straight through. He said that the positive charge and mass in an atom was concentrated in a very small space at the center called the *nucleus*. Electrons revolve around the nucleus.

1.3 Drawbacks

i. It could not explain nuclear collapse. Electrons moving around the nucleus are in accelerated motion. So, an electron will expend energy while revolving and would eventually fall into the nucleus.

ii. Couldn't explain line spectra for various elements.

iii. It could not explain the energies and distribution of electrons around the nucleus.

Important Notations and Definitions

i. General Notation of Elements- ${}^{4}_{2}\mathrm{He}$. Here, 4 denotes Atomic Mass (Protons + Neutrons) and 2 denotes Atomic Number (Protons).

ii. Isotopes- Atoms if the same element having same atomic number (p^{+}), but different mass number(different neutrons). Hydrogen has three Isotopes:

$${}^{1}_{1}\mathrm{H}\text{- Protium, } {}^{2}_{1}\mathrm{H}\text{- Deuterium and } {}^{3}_{1}\mathrm{H}\text{- Tritium.}$$

iii. Isobars- Atoms of different elements having the same mass number. For example: Argon (${}^{40}_{18}\mathrm{Ar}$), Potassium (${}^{40}_{19}\mathrm{K}$) and Calcium (${}^{40}_{20}\mathrm{Ca}$)

iv. Isotones- Atoms of different elements having same number of neutrons. For example: Oxygen ($n = 8$), Nitrogen ($n = 8$) and Carbon ($n = 8$).

v. Isoelectronic Species- Species having same number of electrons.

vi. Wave- A disturbance (energy) that travels through a medium or space, transporting energy from one location to another.

vii. $1\,\text{Å} = 10^{-10}\,\text{m}$, $1\,\text{nm} = 1 \times 10^{-9}\,\text{m}$

2 Bohr's Model

2.1 Developments leading to Bohr's Model

Two developments played a major role: Dual nature of electromagnetic radiation and experimental results regarding atomic spectra which could be explained only through quantized energy levels in atoms.

2.1.1 Waves

When electrically charged particles move under acceleration or oscillate, they create alternating magnetic and electric fields. They are perpendicular to each other and to the direction of propagation of wave. They are called Electromagnetic Waves.

i. Wavelength (λ) - Distance between two neighbouring crests or troughs of a wave.

ii. Frequency(ν) - Number of of waves which pass through a particular point in one second. S.I. unit - Hertz (Hz)

iii. All electromagnetic waves travel at the speed of light. Speed of light, c:

$$c = 3 \times 10^8\,\text{m}\,\text{s}^{-1}$$

$$c = \nu\lambda$$

iv. Amplitude - Height of the crest or depth of the trough. Higher the amplitude, brighter the light.

$$\nu \propto \frac{1}{\lambda}$$

v. Wavenumber, $\overline{\nu}$:

$$\overline{\nu} = \frac{1}{\lambda}$$

2.1.2 Electromagnetic Spectrum

A spectrum is when you place something in increasing or decreasing order. Placing electromagnetic waves in increasing or decreasing order of Wavelength and Frequency, we get the the Electromagnetic Spectrum.

2.1.3 Particle nature of Electromagnetic Radiation

Phenomenon such as diffraction and interference can be explained only by the wave nature of electromagnetic rays. Classical Physics failed to explain Nature of emission of radiation from hot bodies (Black Body radiation), Photoelectric effect, Variation of heat capacity of solids as a function of temperature and the line spectrum of Hydrogen.

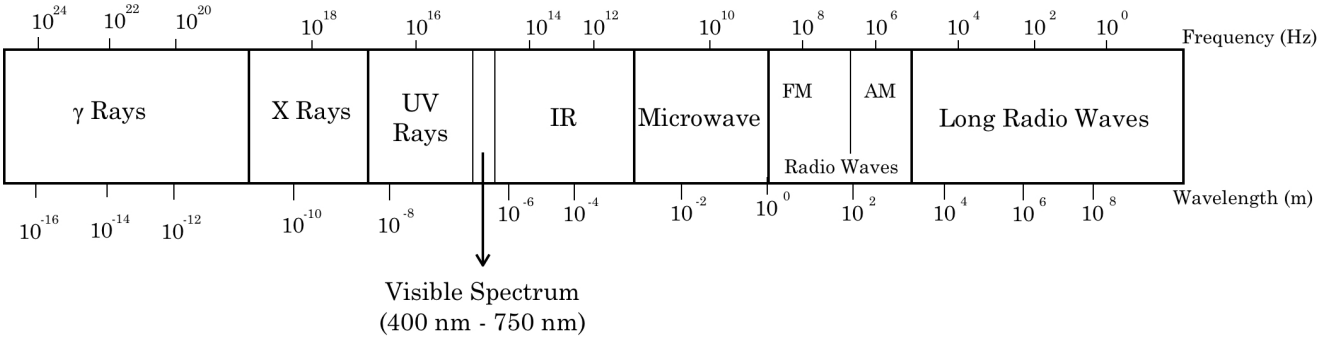


Figure 3: Light Spectrum

2.1.4 Black Body Radiation

The ideal body, which emits and absorbs all frequencies, is called a black body and the radiation emitted by such a body is called black body radiation. BBR has every wavelength in it but intensities of different wavelengths are different.

2.1.5 Planck's Quantum Theory

Atoms and Molecules could emit or absorb energy only in discrete quantities called packets and not in a continuous manner. "Quanta" (plural- "Quantum") is the smallest quantity of energy that can be emitted or absorbed. He suggested:

$$E \propto Frequency(\nu)$$

$$E = h\nu$$

Where h is the Planck's constant equal to 6.626×10^{-34}Js and E is the Energy of 1 Quantum. So, energy lost or gained is $n \cdot h\nu$.

$$E = \frac{hc}{\lambda}$$

1 electron Volt = 1 eV = $1.6 \times 10^{-19}\,\text{C} \times 1\,\text{V} = 1.6 \times 10^{-19}\,\text{J}$

2.1.6 Photoelectric Effect

Certain metals such as K, Rubidium were exposed to a beam of light, electrons were ejected. This phenomenon is called Photoelectric Effect and the ejected electrons are called Photoelectrons. The current generated by the movement of these electrons is called photocurrent. The minimum energy required to remove an electron is called Threshold energy or Work function. So, taking E as the energy provided by photons, E_0 as the threshold energy, we get the remaining energy as Kinetic Energy:

$$E = E_0 + KE$$

$$h\nu = h\nu_0 + \frac{1}{2}mv^2$$

So, *No. of incident rays* $\propto$ *No. of Photoelectrons*. So, $\nu > \nu_0$. As we increase the frequency of incident rays, Kinetic energy increases but not the work function. To increase work function, you must increase the no. of incident rays. For each metal, there is a characteristic minimum frequency ν_0, below which photoelectric effect is not observed.

2.1.7 Stopping Potential

To slow down an ejected electron, we apply a negative potential. The minimum negative potential applied to stop the fastest moving photoelectron. Since, Change in Kinetic Energy is the work done,

$$Change\ in\ KE = Work\ Done$$

$$KE = q \times V$$

Where, q is the the charge and V is the stopping potential.

$$V_{stopping} = \frac{KE_{max}}{q}$$

2.1.8 Dual Nature of Electromagnetic Radiation

Light possesses both particle and wave nature. Particle nature is visible when radiation interacts with matter. Wave nature is exhibited when it propagates.

2.2 Bohr's Model for Hydrogen Atom

His postulates include: Electrons revolved around the nucleus in specific permitted circular orbits and in no others. While in these specific orbits, an electron does not radiate energy. These orbits are known as Stationary Energy Levels. An electron can move from one energy level to another by absorbing/emitting required amount of energy. The frequency of radiation absorbed or emitted is given by:

$$\nu = \frac{\Delta E}{h}$$

Angular momentum of an orbit is always quantized and only those orbits are possible whose angular momentum is integral multiple of $\frac{h}{2\pi}$. A property is called "quantized" if you can only take particular fixed values. So,

$$mvr = n \cdot \frac{h}{2\pi}$$

So mvr can be $\frac{h}{2\pi}$, $\frac{h}{\pi}$, $\frac{3h}{2\pi}$.

2.2.1 Calculation of the radius of Bohr's Orbit

Electrostatic Force = Centripetal Force

$$\frac{kq_1q_2}{r^2} = \frac{mv^2}{r}$$

$$v^2 = \frac{KZe^2}{mr} \tag{1}$$

Since Ze is the total charge of the protons in the nucleus and $k = \frac{1}{4\pi\epsilon_0}$. Since, we know that $mvr = n \cdot \frac{h}{2\pi}$, squaring both sides, we get,

$$v^2 = \frac{n^2h^2}{4\pi^2m^2r^2} \tag{2}$$

Equating (1) and (2), we get:

$$\frac{kZe^2}{mr} = \frac{n^2h^2}{4\pi^2m^2r^2}$$

$$r = \frac{n^2h^2}{4\pi^2mkZe^2}$$

i. n = Orbit number

ii. h = Planck's Constant

iii. m = Mass of electron

iv. k = Constant (refer above)

v. e = Charge on electron

Writing the values of the constants, we get radius of the nth orbit as:

$$r_n = 0.529\frac{n^2}{Z}\text{Å}$$

Taking a_o as Bohr's radius, we get:

$$\boxed{r_n = \frac{n^2 a_0}{Z}} \tag{3}$$

This model is only applicable for single electron systems. We also get $r \propto \frac{n^2}{Z}$

2.2.2 Calculation of velocity of an electron in Bohr's orbit

We have:

$$v = \frac{nh}{2\pi mr}$$

Putting r in the above equation,

$$v = \frac{2\pi Ze^2k}{nh} = 2.18 \times 10^6 \text{m}\,\text{s}^{-1}$$

So, $v \propto Z$; $v \propto \frac{1}{n}$. And, time period of revolution is given as:

$$T \propto \frac{n^3}{Z^2}$$

Since, frequency is inversely proportional to time period,

$$F \propto \frac{Z^2}{n^3}$$

2.2.3 Calculating the energy of an electron

The energy of an electron outside the influence of the nucleus (∞), is assumed to be zero (Zero-Energy State). When electron moves and comes under the influence of the nucleus, it does some work and spends its energy in the process, which results in negative energy. It also indicates that the electron is bound to the nucleus.

So, Potential Energy pf electron at a distance 'r' from the nucleus is equal to the work done in moving the electron from infinity to the point, at a distance r from the nucleus.

$$PE = \int_{r=\infty}^{r=r} F \cdot dr$$

Where F = Coulombic Force

$$PE = \int_{r=\infty}^{r=r} \frac{KZe^2}{r^2} \cdot dr = KZe^2 \int_{r=\infty}^{r=r} \frac{1}{r^2} dr$$

$$PE = KZe^2[\frac{-1}{r}]_{r=\infty}^{r=r}$$

$$PE = \frac{-KZe^2}{r}$$

So, total energy is:

$$T.E = PE + KE = \frac{1}{2}mv^2 - \frac{KZe^2}{r}$$

Since Coulombic Force = Centripetal Force

$$\frac{KZe^2}{r^2} = \frac{mv^2}{r}$$

$$mv^2 = \frac{KZe^2}{r}$$

So, Total energy is:

$$TE = \frac{1}{2}\frac{KZe^2}{r} - \frac{KZe^2}{r} = -\frac{KZe^2}{2r}$$

The following relations emerge: $KE = -\frac{PE}{2}$**,** $TE = \frac{PE}{2}$ **and** $KE = -TE$. For total energy,

$$TE = -\frac{KZe^2}{2r} = -\frac{KZe^2}{2n^2h^2} \times 4\pi^2 mZe^2$$

Putting the values of the constants in the above equation, we get:

$$TE = -13.6\frac{Z^2}{n^2}\,\text{eV atom}^{-1}$$

$$\boxed{TE = -2.18 \times 10^{18}\frac{Z^2}{n^2}\,\text{J atom}^{-1}} \tag{4}$$

$$TE \propto \frac{1}{n^2}$$

For H-atom (Z = 1), $E_1 = -13.6$ eV, $E_2 = -3.4$ eV, $E_3 = -1.51$ eV and $E_4 = -0.85$ eV.

2.2.4 Different Orbits

The K shell is the Ground State (lowest energy state), L- 1st excited state, M- 2nd excited state and N is the 3rd excited state. The ionisation energy is the minimum energy required to move an electron from ground state to $n = \infty$. The opposite sign of the ground state energy is the ionisation energy. Excitation energy is the energy required to move an electron from Ground State to an excited state. For Hydrogen:

$$E_2 - E_1 = 10.2\,\text{eV} = h\nu = \frac{hc}{\lambda}$$

2.3 Failures of Bohr's Theory

Couldn't explain the line spectra of multi-electron systems. He couldn't explain the distribution of electrons in the orbits, couldn't provide mathematical support for the quantization of angular momentum, couldn't explain splitting of spectral lines under influence of electric and magnetic field. Against de-Broglie and Heisenberg's Uncertainty Principle.

2.4 Atomic Spectra

Red has the maximum wavelength, therefore, has the minimum deflection. Violet has the least wavelength, therefore, has the maximum deflection.

2.4.1 Emission and Absorption Spectra

Spectrum of radiation emitted by a substance that has absorbed energy is called an emission spectrum. Atoms, molecules or ions that have absorbed energy are said to be excited. Absorption spectrum is the photographic negative of emission spectrum. A continuum of radiation is passed through a certain sample which absorbs radiation of certain wavelengths. The missing wavelengths leave dark spaces in the bright spectrum.

2.4.2 Line Spectrum of Hydrogen

Passing electric discharge through gaseous Hydrogen, the energetically excited Hydrogen atoms emit electromagnetic radiation of discrete frequencies.

$$\boxed{\overline{\nu} = 1.09 \times 10^7 (\frac{1}{n_1^2} - \frac{1}{n_2^2})} \tag{5}$$

Where R is the Rydberg Constant for Hydrogen.

Series	n_1	n_2	Spectral Region
Lyman	1	2, 3..	Ultraviolet
Balmer	2	3, 4..	Visible
Paschen	3	4, 5..	Infrared
Brackett	4	5, 6..	Infrared
Pfund	5	6, 7..	Infrared

2.4.3 Explanation of Line Spectrum of Hydrogen

Radiation is absorbed if an electron moves from lower energy level to higher energy level and it is emitted when electron moves from higher energy level to lower energy level. This is given as:

$$\Delta E = E_f - E_i$$

$$\Delta E = R_H (\frac{1}{n_1^2} - \frac{1}{n_2^2})$$

Where $R_H = 2.18 \times 10^{-18} J$.

$$\boxed{2.18 \times 10^{-18} J (\frac{1}{n_i^2} - \frac{1}{n_f^2})} \tag{6}$$

Frequency associated is given as:

$$\nu = \frac{\Delta E}{h}$$

$$\boxed{3.29 \times 10^{15} (\frac{1}{n_1^2} - \frac{1}{n_2^2}) \text{Hz}} \tag{7}$$

3 Quantum Mechanical Model

3.1 Towards Quantum Mechanical Model

3.1.1 Dual Behaviour of Matter

French Physicist de Broglie proposed that matter, like radiation, should exhibit dual behaviour, i.e. both particle and wavelike properties. His equaton:

$$\boxed{\lambda = \frac{h}{p} = \frac{h}{mv}} \tag{8}$$

Where λ is the de Broglie wavelength, h is Planck's Constant and p is momentum. According to him, every object in motion has a wave character. The wavelengths associated with everyday objects is so small (large masses), that their wave properties cannot be detected. The wavelengths associated with subatomic particles are large (small masses), that λ can be detected.

3.1.2 Heisenberg's Uncertainty Principle

It states that it is impossible to determine simultaneously, the exact position and exact momentum (or velocity) of an electron. It is given as:

$$\Delta x \cdot \Delta p \geq \frac{h}{4\pi}$$

$$\Delta x \cdot \Delta v \cdot m \geq \frac{h}{4\pi}$$

$$\boxed{\Delta v \cdot \Delta x = \frac{h}{4\pi \cdot m}} \tag{9}$$

It rules out the existence of definite paths or trajectories of electrons and other similar particles.

3.2 Quantum Mechanical Model

Classical mechanics does not take into account Wave-particle duality and thus, fails when applied to microscopic objects. For a system (atom, molecule) whose energy does not change with time, Erwin Schrödinger gave the equation:

$$\hat{H}\psi = E\psi$$

Where $\hat{H}$ is the mathematical operator Hamiltonian, E is the total energy of the system and ψ is the wave function. Solution of this equation gives E and ψ.

3.2.1 Hydrogen Atom and the Schrödinger Equation

Solving Schrödinger's Equation for H-atom, the solution gives the possible energy levels the electron can occupy and corresponding wave function(s) (ψ) of the electron associated with each energy level. Quantized energy states and corresponding wave functions are characterised by a set of quantum numbers- Principal Quantum Number (n), Azimuthal Quantum Number (l) and magnetic quantum number (m_l) arise as a natural consequence of the equation.
The wave function corresponding to that energy state contains all information about the electron, it is a mathematical function whose value depends upon the coordinates of the electron in the atom and does not carry any physical meaning.
Such wave functions of Hydrogen or hydrogen-like species with one electron are called Atomic Orbitals. The probability of finding an electron at a point within an atom is proportional to $|\psi|^2$, which is called probability density.

3.2.2 Quantum Numbers

a. Principal Quantum Number (n) - Determines size and energy of the orbital, identifies the shell and is represented by positive integers with values: *n = 1, 2, 3*. Size of orbital increases with increase in n. The number of allowed orbitals also increase and is given by n^2.

b. Azimuthal Quantum Number (l) - Known as orbital angular momentum or subsidiary quantum number, defines the 3D shape of the orbital. $l = 0...(n-1)$. The number of subshells is equal to the value of n.

c. Magnetic Orbital Quantum Number (m_l) - Spatial orientation of the orbital with respect ot standard set of co-ordinate axis. For any subshell, $2l+1$ values of m_l are possible, given be:

$$m_l = -l, -(l-1), -(l-2)...0, 1, (l-2, (l-1), l$$

l	0	1	2	3	4	5
notation	s	p	d	f	g	h
no. of orbitals	1	3	5	7	9	11

d. Electronic Spin (s) - An electron spins around its own axis. Spin angular momentum of the electron (vector) can have two orientations relative to the chosen axis: distinguishable by the spin quantum numbers (m_s) which can take the values of $+\frac{1}{2}$ or $-\frac{1}{2}$, normally represented by $\uparrow$ (spin up) or $\downarrow$ (spin down). An electron cannot hold more than two electrons and they must have opposite spin.

3.2.3 Shapes of Atomic Orbitals